



Vidyavardhini's College of Engineering & Technology

Department of Computer Engineering

Academic Year : 2024-25

Experiment No. 7
Develop a dashboard and reporting tool based on real time social media data
Date of Performance: 20/2/2025
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Aim: Develop a dashboard and reporting tool based on real time social media data.

Objective: To develop a real-time dashboard and reporting tool using Power BI for analyzing social media data to provide actionable insights for businesses.

Theory:

Tableau is a powerful Business Intelligence (BI) tool used to analyze and visualize data for better decision-making within organizations. It offers a wide range of functionalities that help users connect to various data sources, create interactive dashboards, and share insights across teams.

Data models created from Tableau can be used in several ways, including the following:

- Telling stories through charts and data visualizations.
- Examining what-if scenarios within the data.
- Creating reports that answer questions in real time and help with forecasting to make sure departments meet business metrics.

Key Features:

- Data Connection and Transformation: Tableau allows users to connect to multiple data sources, including databases, spreadsheets, and cloud platforms, enabling seamless data integration. It offers robust data transformation capabilities to clean and prepare data for analysis.
- Interactive Data Visualizations: Users can create rich and interactive visualizations such as charts, graphs, maps, and dashboards using Tableau's intuitive drag-and-drop interface. These visuals can be customized and linked to provide dynamic insights.
- Real-Time Analytics: Tableau supports real-time data analytics, enabling users to monitor key metrics, track performance, and detect trends as they happen. This feature is beneficial for making timely decisions and responding to changing business conditions.
- Collaboration and Sharing: Tableau provides collaboration features that allow teams to work together on projects, share insights, and collaborate on dashboards. Users can publish interactive dashboards to Tableau Server or Tableau Online for widespread access.
- Advanced Analytics: Tableau offers advanced analytics capabilities such as predictive analytics, statistical analysis, and what-if scenarios. These functionalities empower users to gain deeper insights, forecast trends, and simulate business scenarios.

Advantages:

- Ease of Use: Tableau's user-friendly interface and intuitive tools make it accessible to both technical and non-technical users, facilitating widespread adoption within organizations.
- Data Integration: Tableau seamlessly integrates with various data sources, applications, and cloud platforms, allowing users to work with diverse data sets and systems.
- Customer Support: Tableau provides comprehensive customer support, including documentation, training resources, and a vibrant user community, ensuring users have assistance when needed.
- Customization: Tableau offers extensive customization options for visuals and dashboards, enabling users to tailor analytics to specific business needs and objectives.



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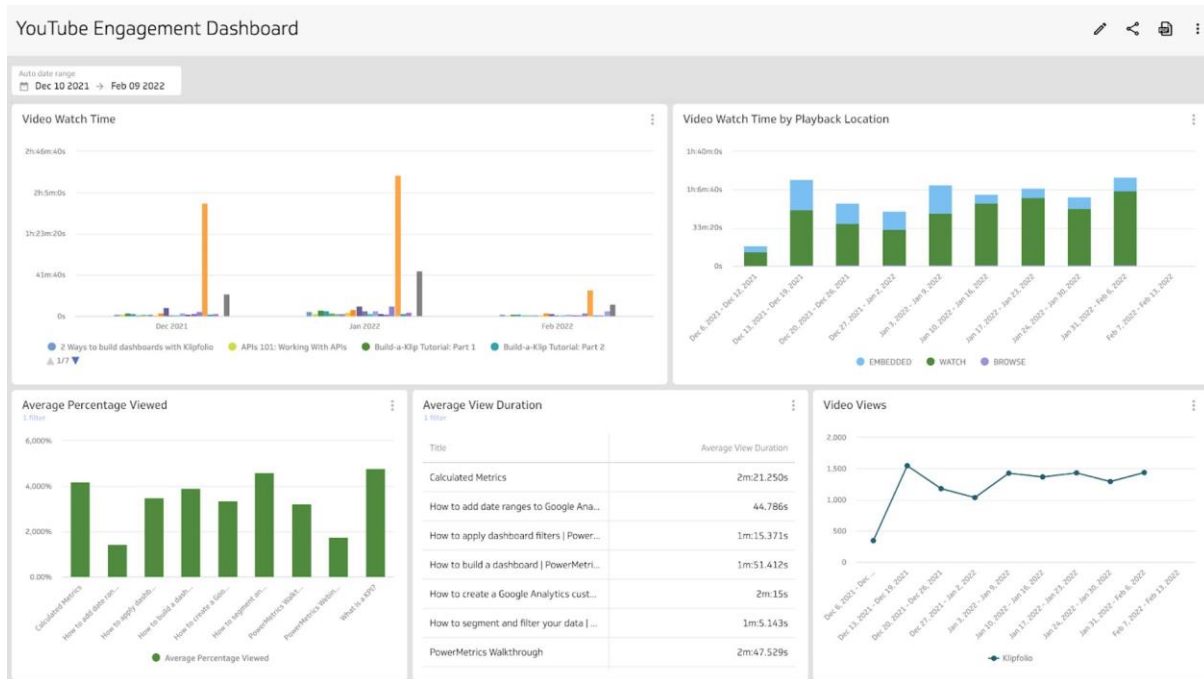
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- **Collaboration and Scalability:** Tableau promotes collaboration among teams through shared workbooks and centralized data repositories. It is scalable and can handle large volumes of data, catering to the needs of enterprises and growing businesses alike.

Dashboard:

1. A YouTube dashboard is a visual representation of the metrics and KPIs that give you insight into your overall engagement and performance on YouTube. You can use the data on your YouTube dashboard to develop your video strategies or identify areas for growth.



Step 1: Define Metrics and KPIs

Before building the dashboard, identify the key metrics you want to track, such as:

- **Video Performance:** Views, likes, comments, shares
- **Engagement Metrics:** Watch time, average view duration, audience retention
- **Subscriber Growth:** New subscribers, total subscribers
- **Traffic Sources:** Where viewers are coming from (search, suggested videos, external sources)
- **Demographics:** Audience age, location, device used



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Step 2: Gather Data from YouTube API

- Use **YouTube Data API v3** to extract real-time data.
- Register on **Google Cloud Console** to obtain API credentials.
- Fetch the required data using API endpoints (e.g., `videos.list`, `channels.list`).
- Store the data in a structured format (e.g., JSON, CSV, or a database).

Step 3: Connect Data to Power BI

- Open Power BI and **connect to the YouTube API** (via Web Connector or a custom script).
- Alternatively, use a **Google Sheets or SQL database** as an intermediary if needed.
- Clean and transform the raw data using **Power Query Editor**.

Step 4: Build Data Model

- Organize tables and establish relationships between them (e.g., videos, channels, engagement metrics).
- Create **calculated columns and measures** using **DAX (Data Analysis Expressions)** for custom insights.
- Optimize performance by reducing redundant data and ensuring efficient queries.

Step 5: Design the Dashboard

- Use **Power BI Visualizations** like:
 - **KPI Cards** (for total views, subscribers, watch time)
 - **Line Charts** (to show trends over time)
 - **Bar Charts** (to compare video performance)
 - **Pie Charts** (to analyze audience demographics)
 - **Tables & Matrices** (for detailed breakdowns)
- Apply filters and slicers for better interactivity.

Step 6: Enable Real-Time Updates

- Configure automatic data refresh for real-time updates.
- Use **Power BI Service** to schedule data refresh from the YouTube API or connected database.

Step 7: Publish and Share the Dashboard

- Publish the Power BI report to **Power BI Service**.
- Share the dashboard with stakeholders by creating **reports and alerts**.
- Embed the dashboard in a website, app, or internal business tool for easy access.

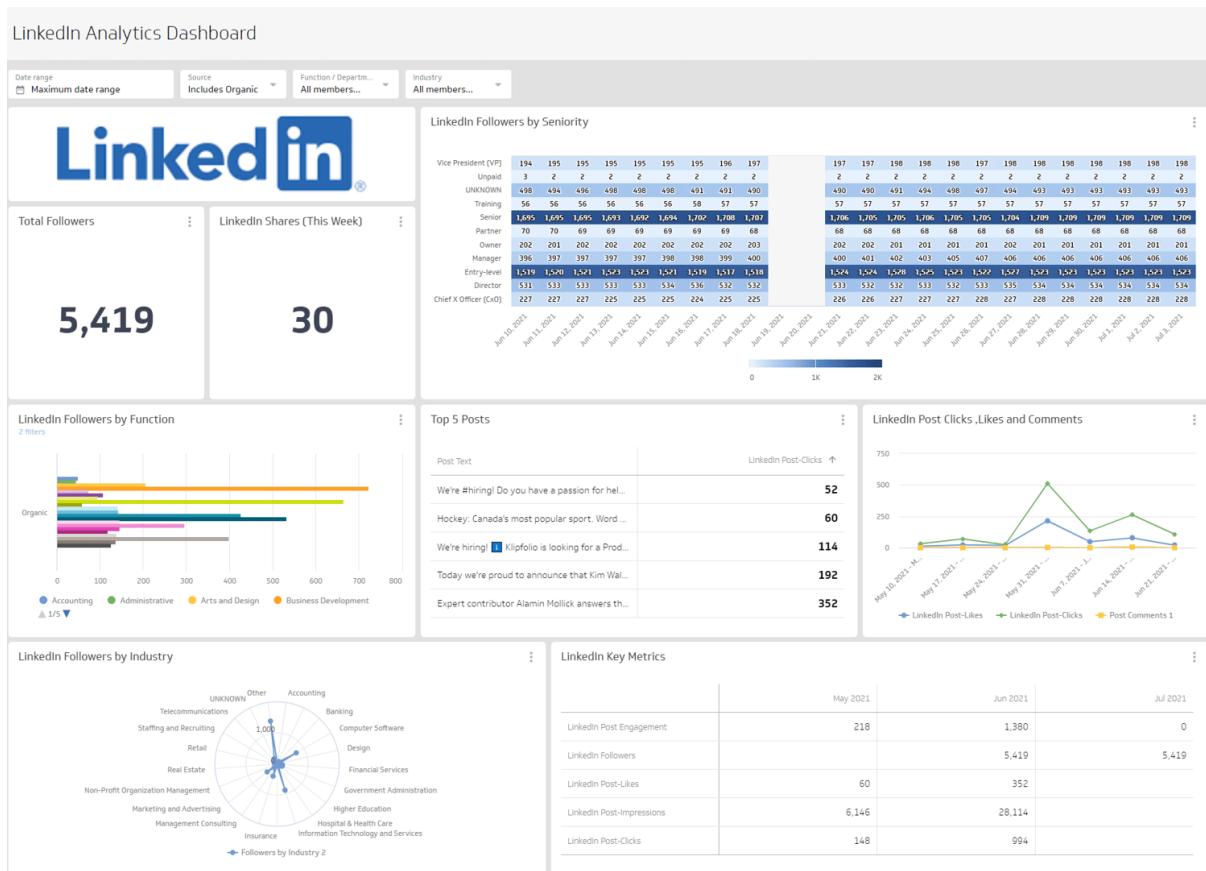


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2. A LinkedIn Pages dashboard displays the metrics and KPIs for your organic social media marketing. Creating a LinkedIn dashboard lets you easily understand how your content performs, if you're reaching your target audience, and the level of engagement with your LinkedIn content.



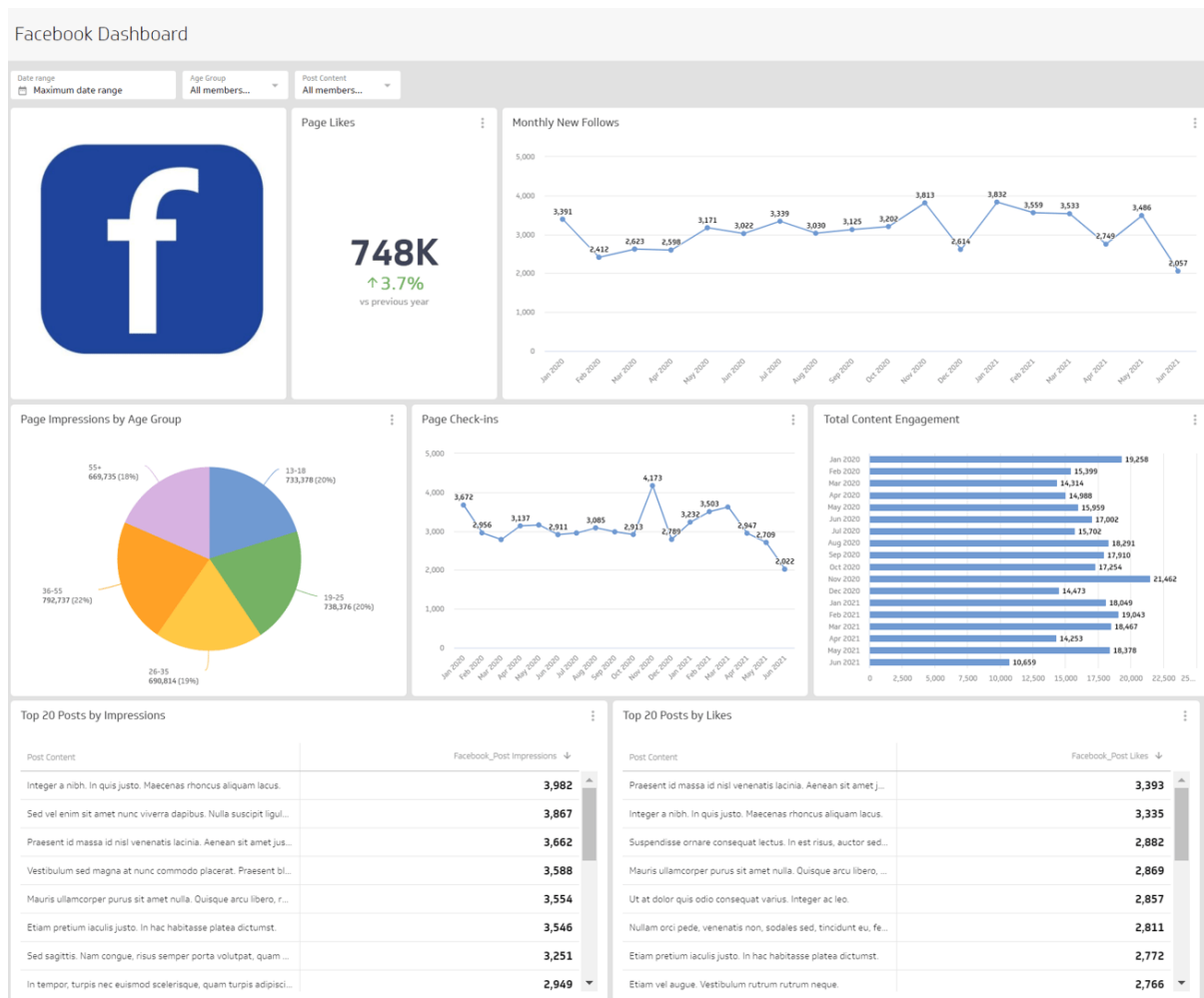


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3. A Facebook Analytics dashboard is a visual reporting tool that displays the metrics and KPIs related to your Facebook Page.



Conclusion: In this experiment, a real-time dashboard and reporting tool was successfully developed using Power BI to analyze social media data. The tool effectively aggregates and visualizes key social media metrics, enabling businesses to gain actionable insights for data-driven decision-making. By leveraging real-time data, the dashboard enhances monitoring capabilities, allowing users to track engagement trends, audience sentiment, and content performance dynamically. The experiment demonstrates the potential of Power BI in transforming raw social media data into meaningful analytics, improving business strategies and marketing effectiveness.